The following exercises represent some of the most important concepts from Math 105L. Students who are beginning Math 106L should be familiar with these concepts, and the problems that are presented here are meant to serve as a review of those concepts. Work together with one partner of your choice to complete the statements or answer the questions. Use a separate piece of paper if you need more room. If you cannot complete this work in class today, arrange to meet your partner before class Friday. Hand in one neatly written copy of this at the beginning of the next class.

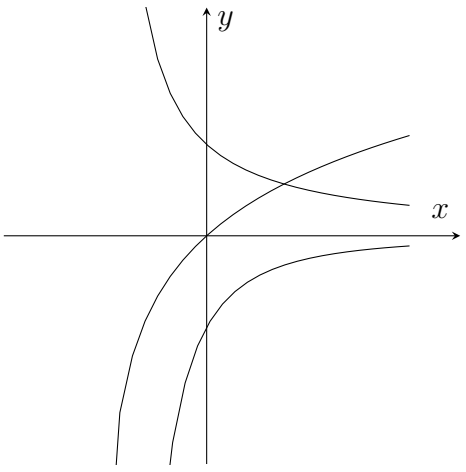
Names: _____ _____

1. The derivative of a function $f(x)$ is defined to be:

$$f'(x) = \lim_{h \to 0}$$

2. Find the linear approximation to the function $f(x) = xe^{kx}$ at $x = 0$. Assuming that $k > 0$, does the linear approximation overestimate or underestimate xe^{kx} near 0? Explain your answer carefully.

3. On the graph below, identify which graph represents $f(x)$, which represents $f'(x)$ and which represents $f''(x)$. Briefly explain your reasoning:



4. Using the table below, estimate $\frac{d}{dx}f(g(x))$ at $x = 1.3$ as closely as possible. Explain your reasoning. (Hint: You will need the chain rule, but you will also need to estimate derivatives. What is the ideal way to estimate them from data?)

x	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
$f(x)$	1.4	1.2	1.1	1.0	1.3	1.4	1.7	1.5	1.6	2.1	2.2
$g(x)$	0.9	1.4	2.2	2.0	1.9	1.5	2.2	3.1	2.5	2.2	2.3

5. Suppose you have collected some data for two variables x and y.

   (a) How can you use straight lines to tell if the data fits $y = Ae^{kx}$?

   (b) How can you use straight lines to tell if the data fits $y = Ax^k$?

6. Suppose that you are watching a rocket take off. You are $1,500$ meters from the launch site. At time t seconds after launch, the height of the rocket in meters is given by $h(t) = 1.39t^2 - 1.65t$. Find the rate at which the distance between you and the rocket is changing at time 10 seconds after launch. You may assume that the rocket rises vertically, and that the ground is flat.

7. A general rational function can be written as

$$f(x) = \frac{a_n x^n + a_{n-1}x^{n-1} + \ldots + a_2 x^2 + a_1 x + a_0}{b_m x^m + b_{m-1}x^{m-1} + \ldots + b_2 x^2 + b_1 x + b_0}.$$

Fill in the following blanks:

- If $m > n$, then $f(x)$ has a horizontal asymptote at ____.
- If m___n, then $f(x)$ has a horizontal asymptote at ____.
(The blanks here should have different answers than in the first bullet point).

8. Fill in the blanks with the words *continuous* and *differentiable*:

- If $f(x)$ is not _____ at $x = a$, then it is not _____ at $x = a$;
- If $f(x)$ is _____ at $x = a$, then it is _____ at $x = a$;
- It is possible for $f(x)$ to be _____ at $x = a$, but not _____ there.

9. Fill in the blanks:

	$f(x) = 10^x$	$f(x) = \log x$
Domain	_____	_____
Range	_____	_____
x intercept	_____	_____
y intercept	_____	_____
Horiz. asymptote	_____	_____
Vert. asymptote	_____	_____

10. (a) $\ln x$ is the inverse of what function?

    (b) Using the derivative of its inverse, show why $\frac{d}{dx} \ln x = \frac{1}{x}$

11. In each of the following, draw a graph of a *differentiable* function $f(x)$ with domain $(-\infty, \infty)$ that satisfies the given conditions. If it is not possible, explain why. Your graph should make it clear what the behavior of $f(x)$ is as x approaches $-\infty$ and ∞.

    (a) $f'(x)$ has three distinct zeroes and $f(x)$ has one zero.

    (b) $f(x)$ has three distinct zeroes and $f'(x)$ has one zero.

12. Without using a calculator, find the the global max and min of $f(x) = x^2 e^{2x}$ on $[0, \infty)$.